

Environmental Impact Dashboard MVP

Overview

The goal is to build a full-stack web application for viewing and analyzing the impact profile of portfolio companies. The platform is designed for investment professionals who need to assess and compare the social and environmental footprint of their investments.

Users manage portfolios containing multiple companies. Each company has impact data from two models:

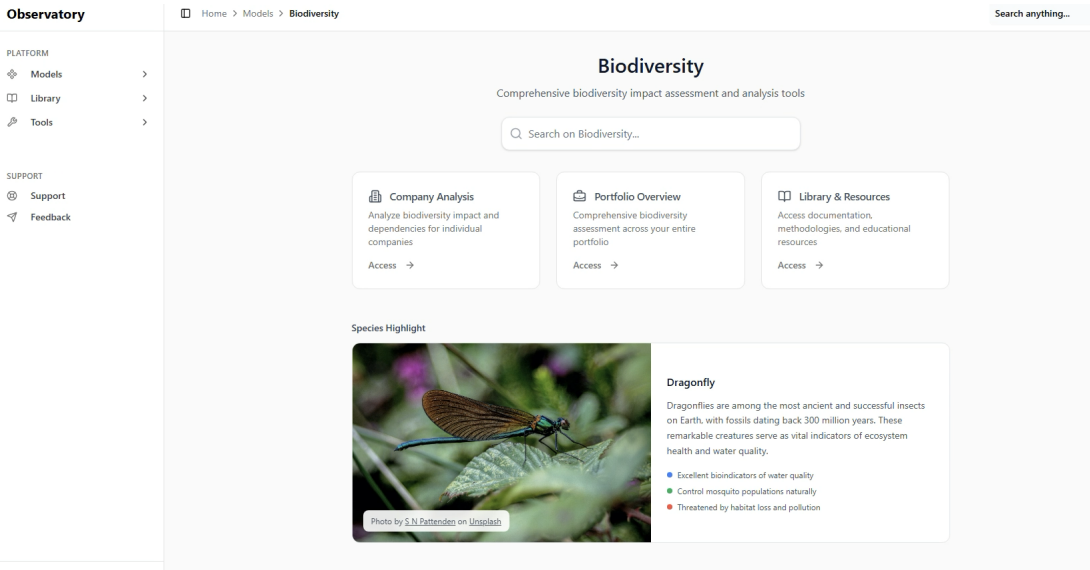
- 1. **Social Impact Model** — measures a company's contribution to human wellbeing in WELLBYs (Wellbeing-Adjusted Life Years). Impact is broken down across 3 scopes (upstream supply chain, own operations, downstream products/services), 5 stakeholder groups (customers, employees, government/communities, shareholders, suppliers), and 12 impact categories (Health, Employment, Energy, Connectivity, etc.). Values can be positive (e.g. employment creation) or negative (e.g. health damage). Currently data is quite sparse — companies only have scores in categories relevant to their business (in the future this will improve). Ideally when thinking about different charts try always use 12 categories, 3 scopes and 5 stakeholder groups even if some may have no data.
- 2. **Biodiversity Impact Model** — measures a company's biodiversity footprint in PDF·yr (Potentially Disappeared Fraction of species × years). Impact is broken down across 3 scopes (direct, upstream, downstream) and 5 environmental categories (Climate change, Water stress, Land use, Eutrophication, Acidification). All values are negative (biodiversity loss). Every company has all 15 scope×category combinations.

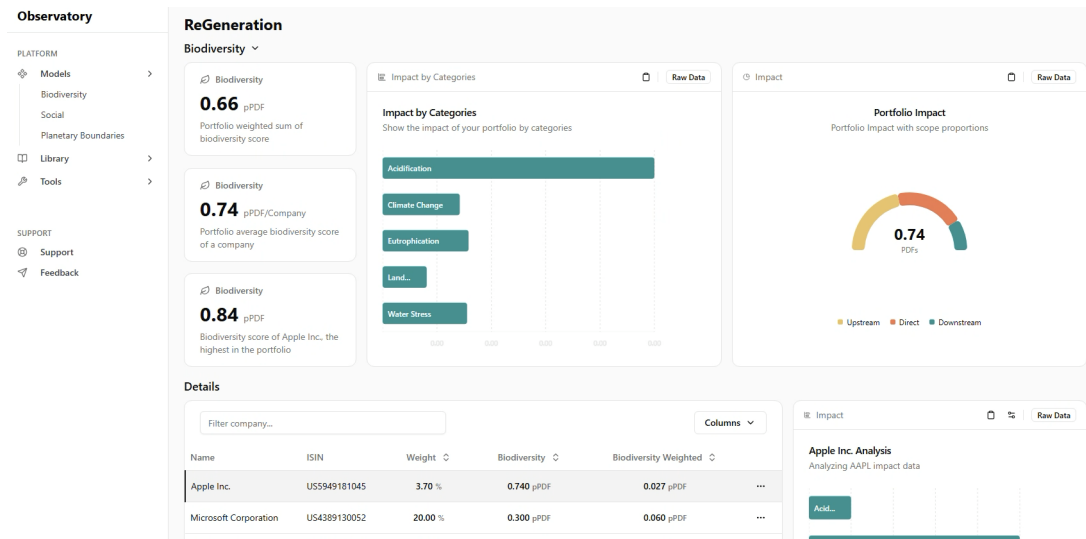
The platform should allow users to quickly navigate between portfolios, drill down into individual companies, and understand the combined social and environmental impact profile of their holdings.

Note: This assignment is intentionally ambitious in scope. We don't expect you to complete everything — the goal is to see how far you can get, how you prioritize, and the quality of what you deliver. Focus on building a solid foundation and making thoughtful trade-offs. You are free to use AI coding agents (Cursor, Claude Code, GitHub Copilot, etc.). However, if you find this assignment too easy, feel free to extend it by adding additional functionality.

Layout/Design

- There is no specific requirement for the layout. The layout should be simple, easy to navigate and understand. See the attached screenshots for design direction — you don't need to follow them exactly, but they can serve as inspiration.





Data

Four CSV files are provided in [Data/](#):

[port_holdings_combined.csv](#) — Portfolio holdings across 3 ETFs (311 rows, 288 unique companies)

Column	Description
portfolio	ETF name (AI & Automation, AI-Enhanced International Equities, Cleaner Planet)
ticker, isin, holdings, cusip, sedol	Company identifiers
pct_of_fund	Portfolio weight (sums to ~100% per portfolio)
shares	Number of shares held
market_value	Market value of the position

[financial_data.csv](#) (optional) — Company financials (288 rows, one per company). Not required for the core functionality, but might be good to have.

Column	Description
ticker, company_name, isin	Company identifiers
market_cap_usd_m	Market capitalization in USD millions
sales_usd_m	Annual revenue in USD millions

[social_impact_model_output.csv](#) — Social impact scores in WELLBYs (3,171 rows, 288 companies)

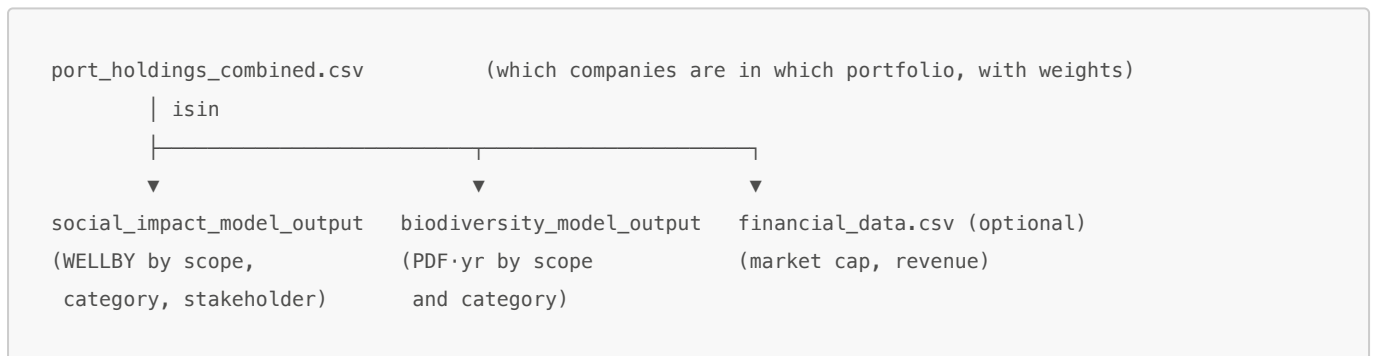
Column	Description
ticker, company_name, isin	Company identifiers
scope	upstream (supply chain), own_ops (direct operations), downstream (products/services)
category	12 categories: connectivity, employment, energy, health, housing, income_wealth, knowledge, leisure, mobility, nutrition, safety, water
stakeholder	customers , employees , gov_communities , shareholders , suppliers
wellby_per_dollar	WELLBY intensity per USD of revenue
wellby_abs	Absolute WELLBY impact

[biodiversity_model_output.csv](#) — Biodiversity footprint in PDF·yr (4,320 rows, 288 companies)

Column	Description
ticker, company_name, isin	Company identifiers
scope	direct , upstream (supply chain), downstream (products/services)
category	5 categories: acidification, climate_change, eutrophication, land_use, water_stress
value	Biodiversity impact intensity (PDF·yr per USD of revenue)

How the data fits together

All four files are linked via **isin** (International Securities Identification Number):



- **Portfolio view:** portfolio → holdings → impact data (joined via **isin**). Portfolio-level scores are calculated by multiplying each company's weight (**pct_of_fund**) by its impact score.
- **Company view:** company impact data (joined via **isin**)
- Both model outputs are in long format (one row per scope/category combination). Aggregate or pivot as needed for display.
- Social impact data is sparse (not every company has every category), while biodiversity data is dense (every company has all 15 scope×category combinations).

Backend

REST API with database seeded from the provided CSVs. Example endpoints:

- **GET /portfolios** — list all portfolios
- **GET /portfolios/:id** — portfolio detail with holdings and aggregated impact summary
- **GET /portfolios/:id/companies** — companies in a portfolio with weights and impact scores
- **GET /companies/:id** — company detail with full impact breakdown (both models)
- **GET /companies/:id/social-impact** — social impact by scope, category, stakeholder
- **GET /companies/:id/biodiversity** — biodiversity impact by scope and category

These are suggestions — feel free to design the API as you see fit.

Tech

- **Frontend (suggested):** React, TypeScript
- **Backend:** Python (preferably FastAPI)
- **Database:** SQLite or PostgreSQL

Submission

GitHub repo with README (setup how to run).